

SOO JEONG CHOI

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PROFESSIONAL APPOINTMENTS

Korea Transportation Institute (KOTI), South Korea
Researcher, Department of AI Mobility Technology

07/2026 – present

EDUCATION

Kyungpook National University, South Korea

03/2023 – 02/2025

M.S. in Geography

Thesis: What makes for an active aging society? A focus on out-of-home activity participation

Advisor: Jae Hyun Lee

Kyungpook National University, South Korea

03/2018 – 02/2023

B.S. in Geography, Summa Cum Laude

RESEARCH INTEREST

Activity-Travel Behavior

· Mobility Equity

· Climate Resilience

Spatio-Temporal Geography

· Sustainable Mobility

PUBLICATIONS

Manuscripts in Preparation

1. **Choi, S.J.**, Jung, Y., Shim J., Jang, D, Lee, J.H., Developing Spatial Multi-Dimensional Mobility Indicators for Individual Travel Behavior Analysis.

Manuscripts Under Review

1. Lee, J.H., **Choi, S.J.**, Debapriya Chakraborty, Scott Hardman, Gil Tal. (2026). Household sharing of electric vehicles and the influence on electric miles travelled. *Humanities and Social Sciences Communications*. (Under Review)
2. **Choi, S.J.**, Su, R, Goulías, K.G., Won, M., Lee J.H. Exploring heterogeneity in older adults' time-of-day activity-travel behavior and its influencing factors: A sequence regression tree analysis in South Korea. *Journal of Transport Geography*. (Under Review)

Peer-Reviewed Publications

1. Kim, M.J., **Choi, S.J.**, Lee, J.H. (2025). The Crisis of Industrial Cities and Industrial

- Patriarchy: A Focus on Special Industrial Crisis Response Regions. *Journal of the Korean Urban Geographical Society*, 28(2), 113-130. doi: 10.21189/JKUGS.28.2.8
2. Won, M.S., Shim, J., **Choi, S.J.**, Lee, J.H. (2024). Development of Multi-Dimensional Mobility Indicators (MDMI) for Agent-based Travel Pattern Classification. *Journal of Korean Society of Transportation*, 42(6), 674-693. doi: 10.7470/jkst.2024.42.6.674

RESEARCH EXPERIENCE

Development of next-generation AI integrated mobility simulation and prediction /application technologies for metropolitan cities

- Funder: Ministry of Science and ICT
- Project Duration: Jul. 2024 – Dec. 2027 (Participated: Jul. 2026. – Present)
- Funding: 6,825 million KRW (\$4.9 million (USD))
- Role: Researcher (at Korea Transport Institute)

National Highway Network - GIS Policy Impact Analysis (2025)

- Funder: Korea Research Institute for Human Settlements
- Project Duration: Sep. 2025 – Dec. 2025 (Participated: Sep. 2025 – Dec. 2026)
- Role: Research Assistant (at Kyungpook National University)

Multi-Dimensional Mobility Index Mapping

- Funder: Korea Transport Institute
- Project Duration: Jul. 2025 – Feb. 2026 (Participated: Jul. 2025 – Feb. 2026)
- Role: Research Assistant (at Kyungpook National University)

Carbon-Neutral Priority-Zone Integrated Scenario Analysis

- Funder: Korea Research Institute for Human Settlements
- Project Duration: May. 2025 – Oct. 2025 (Participated: May. 2025 – Oct. 2025)
- Role: Research Assistant (at Kyungpook National University)

Network Analysis and Database for Mobility Assessment

- Funder: Korea Transport Institute
- Project Duration: Apr. 2025 – Oct. 2025 (Participated: Apr. 2025 – Oct. 2025)
- Role: Research Assistant (at Kyungpook National University)

National Highway Network - GIS Policy Impact Analysis (2024)

- Funder: Korea Research Institute for Human Settlements
- Project Duration: Sep. 2024 – Dec. 2024 (Participated: Sep. 2024 – Dec. 2024)
- Role: Research Assistant (at Kyungpook National University)

Spatial Strategy for Carbon-Neutral Energy Sector

- Funder: Korea Research Institute for Human Settlements
- Project Duration: May. 2024 – Oct. 2024 (Participated: May. 2024 – Oct. 2024)
- Role: Research Assistant (at Kyungpook National University)

Grid-level EV Charging Infrastructure Siting

- Funder: Korea Research Institute for Human Settlements

- Project Duration: Mar. 2023 – Aug. 2023 (Participated: Mar. 2023 – Aug. 2023)
- Role: Research Assistant (at Kyungpook National University)

Smartphone Activity Data - Social-Distancing Policy Impact

- Funder: Korea Research Institute for Human Settlements
- Project Duration: Mar. 2023 – Dec. 2023 (Participated: Mar. 2023 – Dec. 2023)
- Role: Research Assistant (at Kyungpook National University)

Developing a Methodological Framework for Analyzing Mobility Patterns among Urban and Intercity Rail Users

- Funder: Korea Transport Institute
- Project Duration: Jan. 2023 – Apr. 2023 (Participated: Jan. 2023 – Apr. 2023)
- Role: Research Assistant (at Kyungpook National University)

Mid- to Long-Term Mobility Pattern Change with Participatory Geographic Information

- Funder: National Research Foundation of Korea
- Project Duration: Sep. 2022 – Feb. 2025 (Participated: Jan. 2023 – Feb. 2025)
- Role: Research Assistant (at Kyungpook National University)

TEACHING EXPERIENCE

Teaching Assistant (KNU)

Transport Geography

Fall 2025

- Prepared lecture materials covering transportation networks, distance and cost, mobility and accessibility, travel and activity behavior, and transport policy, supporting students' understanding of transport geography from both the infrastructural and spatial-relational perspectives.
- Evaluated student assignments and research reports and held regular office hours to provide individualized feedback and guidance on course concepts and analytical methods.

Geographical Field Survey

Spring 2024

- Contributed to the development of lecture materials incorporating quantitative and qualitative geographic research methodologies to facilitate effective exploration of field study areas.
- Took full responsibility for the overall preparation required for geographic field trips.

Introduction to Geographic Information System and Science

Spring 2023

- Developed and curated comprehensive set of laboratory materials and assignments, guiding undergraduate students through essential analytic techniques for using geographic data.
- Conducted all practical laboratory sessions and the primary software used for practical exercises was QGIS.

PRESENTATIONS

1. Exploring Urban Mobility Patterns with Multi-Dimensional Spatial Characteristics Sequence

- Analysis. *GISER Symposium, 2026 AAG (American Association of Geographers) Annual Meeting*. San Francisco, CA. March 19, 2026. (Oral presentation)
2. Fast-Aging Society's Elderly Travel Behavior Analysis with Sequence Tree Method. *17th IATBR (International Association for Travel Behavior Research) Conference*. Vienna, Austria. July 18, 2024. (**Peer-reviewed extended abstract**; Oral presentation)
 3. Which spatial components affect Active aging society?: Focused on activity-travel behavior of older people, social infrastructure, and public transportation. *2024 Annual Conference of Korean Geographical Society*. Seoul, South Korea. June 28, 2024. (Oral presentation; **Award for Outstanding Oral Presentation**)
 4. Empirical Analysis of the Relationships between Land Use and Activity-Travel Behavior: Sequential Analysis Approaches. *2024 AAG (American Association of Geographers) Annual Meeting*. Honolulu, HI. April 19, 2024. (Poster presentation)
 5. Comparative Analysis of Location-based Land use sequence patterns and Activity-Travel sequence patterns. *The 90th Conference of Korean Society of Transportation*. Seoul, South Korea. March 28, 2024. (Poster presentation)
 6. Understanding Daily Activity-Travel Patterns of Elderly Population in South Korea Using Tree-Structured Analysis of Sequence. *The 15th Korea-China-Japan Joint Conference on Geography*. Seoul, South Korea. October 22, 2023. (Oral presentation)
 7. An Explanatory analysis of applicability of Motif for mobility indicator: Based on Motif Composition ratio of Socioeconomic variables. *Joint Conference of Korean Society of Transportation & Korea Planning Association*. Busan, South Korea. October 12, 2023 (Oral presentation)

SCOLARSHIPS & AWARDS

Kyungpook National University Development Project Research Fund	2024-2025
Hackathon for Improving Demand-Responsive Transportation Services for Mobility-Impaired Individuals Using Public Data (2nd prize)	2023
KNU Graduate student Global challenger program	2023
Honors Club in KNU (for Undergraduate students)	2023
Scholarship for Academic Excellence, KNU	2020
Mentoring Program (for teenagers) Scholarship, KNU	2019
Scholarship for Academic Excellence, KNU	2018

SKILLS & OTHERS

- Programming: R, Python, SQL (Big Query)
- Tools: QGIS, ArcGIS, GeoDa, Google Earth Engine, SUMO, MS Office
- Languages: Korean (Native), English (Advanced)